

Deliverable 7 – Power BI Data Modeling Report

Healthcare Patient Experience Analytics using Power BI

Introduction

A well-designed data model is the foundation of an efficient Business Intelligence solution. In this project, a **Star Schema** data model was implemented in Microsoft Power BI to organize healthcare survey data into fact and dimension tables. This approach improves report performance, simplifies DAX calculations, reduces data redundancy, and supports scalable analytics.

The model enables users to analyze patient satisfaction across different states, regions, reporting periods, healthcare quality measures, and survey response categories while ensuring consistent and reliable reporting.

Data Modeling Approach

The project follows the **Star Schema** modeling technique.

In a Star Schema:

- Fact tables store measurable business data.
- Dimension tables provide descriptive information used for filtering and categorization.
- Dimension tables are connected to fact tables through one-to-many relationships.
- The model minimizes redundancy and improves query performance.

This design provides a scalable and efficient foundation for healthcare analytics.

Star Schema

Fact Tables

Fact Table	Description
National Results	Stores national-level patient satisfaction results used for benchmarking and KPI calculations.
State Results	Stores patient satisfaction results for each participating state.
Responses	Stores survey response categories and percentage values used for KPI calculations.

Dimension Tables

Dimension Table	Description
Measures	Contains healthcare quality measures evaluated in the HCAHPS survey.
Questions	Contains survey questions associated with each healthcare measure.
Reports	Stores reporting period information for trend analysis.
States	Stores state names, state codes, and regional classifications.

Relationships

The following relationships were created within the Power BI data model.

Dimension Table	Fact Table	Relationship
Measures	National Results	One-to-Many
Measures	State Results	One-to-Many
Questions	National Results	One-to-Many
Questions	State Results	One-to-Many
Reports	National Results	One-to-Many
Reports	State Results	One-to-Many
States	State Results	One-to-Many
Responses	National Results	One-to-Many
Responses	State Results	One-to-Many

These relationships ensure accurate aggregation and filtering throughout the dashboard.

Cardinality

The following cardinality settings were implemented.

Relationship	Cardinality
Dimension → Fact	One-to-Many (1:*)

Explanation

Each dimension table contains unique records, while each fact table contains multiple transactional or survey records related to those dimensions. Therefore, a **One-to-Many** relationship is appropriate for all dimension-to-fact relationships.

Cross-filter Direction

The project uses **Single Cross-filter Direction** for all relationships.

Reason

Single-direction filtering provides several advantages:

- Improves report performance.
 - Prevents ambiguous relationships.
 - Simplifies DAX calculations.
 - Reduces the possibility of circular dependencies.
 - Ensures predictable filtering behavior throughout the dashboards.
-

Modeling Decisions

The following design decisions were made while building the data model:

1. Star Schema Design

A Star Schema was implemented to improve scalability, maintainability, and analytical performance.

2. Fact and Dimension Separation

Fact tables were separated from dimension tables to reduce redundancy and simplify reporting.

3. Primary and Foreign Keys

Relationships were created using unique key columns to maintain data integrity and ensure accurate filtering.

4. Single Direction Relationships

Single-direction filtering was used to improve performance and avoid ambiguous filtering paths.

5. Data Type Standardization

All key columns were assigned appropriate data types before relationships were created.

6. Optimized DAX Performance

The model was designed to minimize unnecessary calculations and improve DAX execution speed.

7. Business-Friendly Structure

Table names, column names, and relationships were organized using meaningful business terminology to improve usability.

Benefits of the Data Model

The implemented data model provides the following benefits:

- Faster dashboard performance.
 - Simplified DAX calculations.
 - Reduced data redundancy.
 - Improved scalability.
 - Consistent KPI calculations.
 - Accurate filtering across reports.
 - Easy maintenance and future enhancements.
 - Reliable business reporting.
-

Data Model Summary

Component	Count
Fact Tables	3
Dimension Tables	4
Total Tables	7

Component	Count
Relationship Type	One-to-Many
Cross-filter Direction	Single
Modeling Technique	Star Schema

Conclusion

The Power BI data model was designed using a Star Schema architecture to support efficient healthcare analytics and reporting. By separating fact and dimension tables and implementing one-to-many relationships with single-direction filtering, the model provides a scalable, high-performance foundation for interactive dashboards and KPI calculations. This design ensures accurate analysis of patient satisfaction across multiple healthcare dimensions and supports informed decision-making for healthcare stakeholders.

